Visual Language Models and Robotics: A Survey of the Transition from Teleoperation to Autonomy

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Abstract

Recent advances in visual language models (VLMs) and robot imitation learning are catalyzing a fundamental transformation in robotics, enabling the transition from teleoperated systems to autonomous robots capable of perceiving, understanding, and acting in the physical world with unprecedented flexibility. This survey examines the current state and future prospects of VLM-powered robotics across multiple domains, including industrial applications, healthcare, logistics, and consumer robotics. We analyze how large multimodal AI systems combine vision and language understanding with control policies, allowing robots to learn from human demonstrations, follow natural language instructions, and generalize knowledge to new situations. The survey covers key technological trends including open-source robotics frameworks, simulation-based learning, and the integration of large language models as high-level planners. Our analysis reveals that while significant progress has been made in constrained environments, the path to general-purpose autonomous robots requires continued advances in multimodal foundation models, safety mechanisms, and human-robot interaction paradigms.

1 Introduction

The field of robotics is experiencing a paradigm shift driven by the convergence of visual language models (VLMs), imitation learning, and large-scale multimodal AI systems. Traditional robotic systems have long relied on explicit programming for specific tasks or teleoperation for complex scenarios requiring human expertise. However, recent breakthroughs in artificial intelligence are enabling a new generation of robots that can learn from demonstrations, understand natural language instructions, and adapt to novel situations [1], [2].

This transformation is particularly evident in the development of systems like Hugging Face's LeRobot platform, which provides end-to-end learning capabilities for robotics applications [1]. These platforms demonstrate how modern AI can bridge the gap between teleoperated machines—robots remotely controlled by humans—and autonomous robots capable of performing tasks with minimal direct oversight [3].

The implications of this technological evolution extend across multiple domains:

- **Industrial applications**: Where robots are transitioning from rigid automation to flexible, adaptive systems capable of handling high-mix manufacturing and complex assembly tasks. Companies like GrayMatter Robotics demonstrate how AI can eliminate manual robot programming, allowing operators to describe processes in natural language [4]

- **Healthcare and medical robotics**: Where surgical robots are learning to perform procedures by observing human demonstrations, potentially reducing surgeon fatigue while increasing precision. Johns Hopkins University's breakthrough in training surgical robots through video observation exemplifies this transformation [5]

- **Logistics and transportation**: Where autonomous systems are scaling from controlled environments to dynamic, real-world scenarios. Tesla's Full Self-Driving system and Waymo's autonomous trucking operations demonstrate practical deployment of vision-based autonomy [6], [7]

- **Consumer robotics**: Where household robots are gaining commonsense reasoning capabilities through internet-scale vision-language training. iRobot's latest Roomba models incorporate AI-powered object recognition, while research prototypes demonstrate understanding of complex natural language commands [2], [8] This survey provides a comprehensive analysis of how visual language models are reshaping robotics, examining both current achievements and future prospects. We explore the technical foundations enabling this transformation, analyze domain-specific applications, and identify key trends that will shape the next generation of autonomous robotic systems.

2 Technical Background

2.1 Visual Language Models in Robotics

Visual Language Models represent a significant advancement in multimodal AI, combining computer vision and natural language processing to enable robots to understand and reason about their environment in ways that mirror human cognition [9]. These models are trained on large-scale datasets containing paired visual and textual information, allowing them to develop rich representations that connect visual observations with semantic understanding.

The architecture of VLMs for robotics typically consists of several key components:

- **Vision encoder**: Processes visual input from cameras and sensors to extract meaningful features from the robot's environment. Modern implementations often build upon CLIP-style architectures [10] or specialized visual transformers designed for robotic perception
- **Language encoder**: Handles natural language instructions and contextual information, enabling human-robot communication. GPT-4 and similar large language models serve as foundation architectures for understanding complex commands [11]
- **Multimodal fusion**: Combines visual and linguistic representations to create unified understanding of tasks and objectives. Google's Flamingo model demonstrates effective cross-modal attention mechanisms for this integration [12]
- **Action decoder**: Translates high-level understanding into specific motor commands and control signals for robot actuators. RT-1 and RT-2 showcase how transformer architectures can directly output robot actions [2], [13]

Google DeepMind's RT-2 (Robotics Transformer 2) exemplifies this approach, demonstrating how robots can leverage web-scale vision-language training to perform novel tasks without explicit programming [2]. By combining internet-scale visual knowledge with physical actions, RT-2 enables robots to exhibit emergent capabilities such as identifying novel objects or using tools in creative ways.

2.2 Imitation Learning and Human Demonstration

Imitation learning has emerged as a crucial paradigm for robot skill acquisition, allowing systems to learn complex behaviors by observing human demonstrations rather than through exhaustive programming or trial-and-error exploration [14]. This approach is particularly valuable for tasks that are difficult to specify explicitly but can be easily demonstrated by human experts.

The process typically involves several stages:

- **Data collection**: Recording human demonstrations through teleoperation, motion capture, or video observation. The da Vinci surgical system's teleoperation data provides rich examples for surgical robotics training [15]

- **Behavior cloning**: Training neural networks to map observations to actions based on demonstrated behaviors. This approach has been successfully applied in autonomous driving, where millions of human driving hours inform vehicle control policies [6]

- **Inverse reinforcement learning**: Inferring the underlying objectives and reward structures from human demonstrations. Google's SayCan project demonstrates how robots can learn task preferences by observing human behavior patterns [16]

- **Policy refinement**: Improving learned behaviors through additional training or fine-tuning in the target environment. NVIDIA's Isaac Sim platform enables safe policy refinement through high-fidelity simulation before real-world deployment [17]

Recent advances have shown remarkable success in applying imitation learning to surgical robotics, where researchers at Johns Hopkins University trained robots to perform suturing and other delicate procedures by learning directly from surgical videos [5]. These systems achieved performance levels comparable to human surgeons without explicit programming of individual movements.

2.3 Foundation Models and Multimodal AI

The emergence of foundation models—large-scale AI systems trained on diverse datasets—has fundamentally changed the landscape of robotic intelligence [18]. These models provide a strong basis for generalization, enabling robots to apply knowledge learned in one context to novel situations and tasks.

Key characteristics of foundation models for robotics include:

- **Scale**: Training on massive datasets encompassing diverse visual, textual, and behavioral data
- **Generalization**: Ability to transfer knowledge across different tasks, environments, and robot embodiments
- **Emergence**: Development of capabilities not explicitly programmed, arising from the complexity of the training data and model architecture
- **Adaptability**: Capacity for fine-tuning and customization for specific applications and domains

3 Industrial Applications

3.1 Manufacturing and Assembly

The manufacturing sector is witnessing a significant transformation as visual language models enable more flexible and adaptive robotic systems. Traditional industrial robots required extensive programming for each new product or task, limiting automation primarily to mass production scenarios. Modern AI-powered systems are changing this paradigm by enabling robots to learn new tasks through demonstration and natural language instruction [19].

GrayMatter Robotics exemplifies this transformation, demonstrating how generative AI and large language models can eliminate manual robot programming for complex tasks [4]. Their approach allows operators to describe motions or processes in natural language, with AI systems automatically generating the corresponding robot instructions. This capability dramatically reduces the expertise barrier for robot deployment and accelerates the implementation of automation in high-mix manufacturing environments.

Current applications include:

- **Surface finishing operations**: Robots learning sanding, polishing, and painting techniques through imitation learning
- **Assembly tasks**: Adaptive systems capable of handling variations in part geometry and assembly sequences
- **Quality inspection**: Vision-language models enabling robots to identify defects and anomalies based on natural language descriptions
- **Material handling**: Intelligent systems capable of identifying and manipulating diverse objects without explicit programming

3.2 Heavy Equipment and Construction

The construction and heavy equipment industry represents a compelling application domain for the transition from teleoperation to autonomy. Companies like Hive Autonomy are pioneering this transformation through advanced remote and autonomous solutions for industrial machinery [20].

Hive Autonomy's work with partners including Volvo and Yara demonstrates practical implementations of this technology. Their remote-operated wheel loaders can be supervised from central control stations, with plans to scale operations where a single operator controls multiple machines [20]. This approach significantly improves operator safety by removing humans from dangerous work sites while maintaining operational efficiency.

The progression toward autonomy in heavy equipment involves several key technological components:

- **Environmental perception**: Advanced sensor systems and computer vision for navigation and obstacle detection. Companies like Caterpillar integrate LiDAR, radar, and camera systems for comprehensive environmental awareness [21]

- **Predictive control**: AI systems that anticipate and respond to changing conditions in construction environments. Komatsu's autonomous mining trucks demonstrate predictive path planning in dynamic terrains [20]

- **Fleet coordination**: Multi-agent systems enabling coordination between multiple autonomous machines. Hive Autonomy's approach allows single operators to supervise multiple remote-controlled wheel loaders simultaneously [22]

- **Human oversight**: Sophisticated interfaces allowing remote operators to intervene when necessary. Boston Dynamics' Spot robot exemplifies advanced teleoperation interfaces for industrial inspection tasks [23]

3.3 Warehousing and Logistics Operations

Modern warehousing and fulfillment operations are increasingly adopting AI-powered robotic systems that combine visual understanding with flexible manipulation capabilities. These systems represent a significant advancement over traditional automation, which was limited to handling pre-defined objects in structured environments.

The integration of vision-language models enables warehouse robots to:

- **Handle diverse inventory**: Process and sort products without explicit programming for each item type. Amazon's fulfillment centers employ Kiva robots that can identify and manipulate thousands of different products using computer vision [24]

- **Adapt to packaging variations**: Recognize and manipulate items despite differences in size, shape, and packaging. Modern warehouse robots can handle everything from small electronics to large appliances through adaptive grasping strategies [25]

- **Respond to natural language instructions**: Accept task specifications from human operators using conversational interfaces. Voice-directed warehouse systems now enable workers to communicate with robotic systems using natural speech commands

- **Learn from demonstrations**: Acquire new manipulation skills by observing human workers. This approach significantly reduces the time needed to deploy robots for new product lines or packaging formats [14]

The collaboration between Hugging Face and NVIDIA specifically targets these applications, aiming to drive breakthroughs in logistics through shared models and simulation data [26]. This partnership exemplifies the industry trend toward open-source development and collaborative advancement of robotic capabilities.

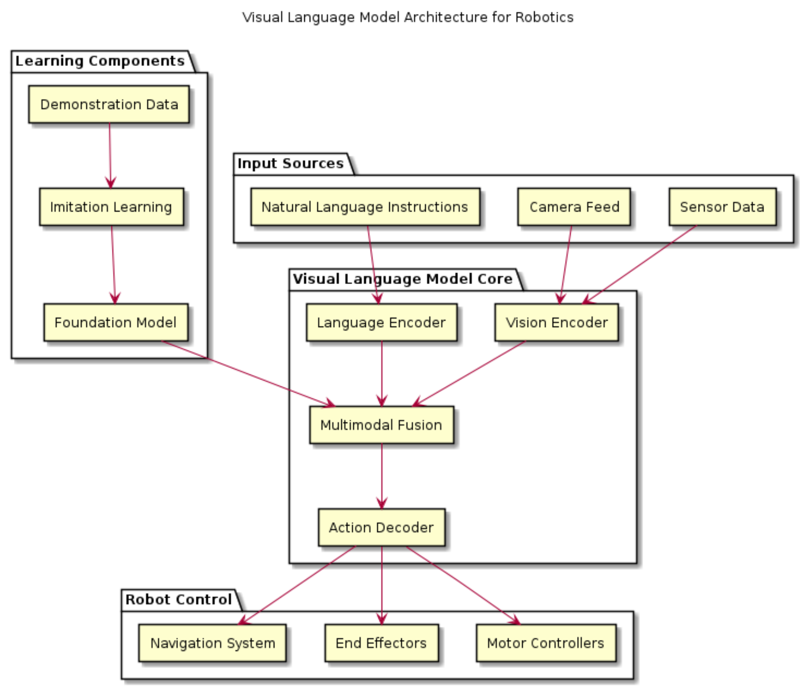

Figure 1: Visual Language Model architecture for robotics showing the integration of vision, language, and action components, based on designs from RT-1 [13] and RT-2 [2]

4 Healthcare and Medical Robotics

4.1 Surgical Robotics and Autonomous Procedures

The healthcare sector is experiencing a revolutionary transformation in surgical robotics, driven by advances in imitation learning and visual language models. Traditional surgical robots, such as the da Vinci system, operate primarily as teleoperated tools requiring direct surgeon control. Recent breakthroughs are enabling a shift toward semi-autonomous surgical assistance that can reduce surgeon fatigue while increasing precision [27].

A landmark development occurred in 2024 when researchers at Johns Hopkins University successfully trained a surgical robot by having it observe videos of human surgeons performing procedures [5]. The robot learned to perform suturing and other delicate maneuvers with skill levels comparable to human doctors, without requiring explicit programming for each movement. This achievement demonstrates the potential for imitation learning to transfer complex manual skills from human experts to robotic systems.

The technical approach involves several key innovations:

- **Video-based learning**: Training systems on large datasets of recorded surgical procedures to extract motion patterns and decision-making strategies. Johns Hopkins University's breakthrough involved training on thousands of hours of surgical videos [5]

- **Instrument tracking**: Computer vision systems that precisely monitor surgical tool positions and movements during procedures. Modern tracking systems achieve sub-millimeter accuracy in real-time instrument localization [27]

- **Skill transfer**: Machine learning algorithms that map observed human actions to appropriate robot motor commands. This process involves understanding both the mechanical constraints of robotic systems and the intent behind human movements [14]

- **Safety integration**: Fail-safe mechanisms that ensure human oversight and intervention capabilities are maintained. Intuitive Surgical's latest da Vinci systems incorporate multiple layers of safety monitoring and automatic stopping mechanisms [28]

Future applications in surgical robotics include semi-autonomous assistance for routine procedures such as suture closure, precise instrument positioning, and even complex surgical sub-tasks performed under human supervision. This evolution promises to reduce surgeon fatigue during lengthy procedures while potentially improving consistency and precision in surgical outcomes.

4.2 Medical Service and Assistive Robotics

Beyond surgical applications, visual language models are enhancing medical service robots and assistive devices throughout healthcare facilities. Traditional telepresence robots allow remote doctors to observe patients but require constant human control for navigation and interaction. Modern AI-powered systems are developing capabilities for autonomous navigation and intelligent assistance [29].

Current developments include:

- **Autonomous hospital navigation**: Robots capable of finding specific patients, equipment, or locations based on natural language requests. TUG robots in hospitals can navigate complex floor plans and elevator systems to deliver medications and supplies autonomously [29]

- **Patient monitoring**: Systems that can recognize and report specific visual indicators or changes in patient condition. Advanced monitoring robots can detect falls, assess mobility, and alert healthcare staff to emergency situations [27]

- **Rehabilitation assistance**: Exoskeletons and therapy robots that learn optimal supportive motions through imitation of physical therapists. ReWalk and similar systems now incorporate AI to adapt assistance patterns to individual patient needs and recovery progress

- **Medication delivery**: Autonomous systems capable of safely transporting and delivering medications throughout healthcare facilities. These systems integrate with hospital information systems to ensure accurate delivery while maintaining chain of custody requirements [24]

The integration of vision-language understanding enables these systems to operate more intelligently in complex hospital environments, interpreting contextual cues and responding appropriately to dynamic situations that would challenge traditional rule-based systems.

5 Consumer and Service Robotics

5.1 Household and Personal Robotics

The consumer robotics sector is beginning to realize the potential of visual language models for creating more capable and intuitive household assistants. While general-purpose humanoid robots remain in development, specialized consumer robots are demonstrating increasingly sophisticated capabilities through VLM integration [30].

Research prototypes have demonstrated robots capable of understanding complex natural language commands such as "bring me an energy drink" and reasoning about which objects satisfy the request based on visual understanding and common-sense knowledge [2]. This represents a fundamental advance over traditional robots that were limited to recognizing only pre-programmed objects or following rigid command structures.

Key capabilities emerging in consumer robotics include:

- **Common-sense reasoning**: Understanding implicit requirements and context in household tasks. Google's RT-2 demonstrates how robots can understand that a "energy drink" request might be satisfied by various beverage types based on contextual understanding [2]
- **Object generalization**: Recognizing and manipulating items not encountered during training. Modern robotic vacuum cleaners can identify and avoid pet toys, cables, and other household objects without explicit programming for each item type [8]
- **Natural language interaction**: Accepting task descriptions in conversational language rather than formal commands. Amazon's Astro robot showcases natural language interaction for home security and communication tasks [30]
- **Adaptive behavior**: Learning and improving task performance through observation and feedback. Home robots can learn family routines, preferred cleaning schedules, and adapt to changing household layouts through continuous observation [31]

Google's Robotics Transformer models (RT-1 and RT-2) exemplify this progress by combining internet-scale vision-language training with robot-specific demonstration data [13]. This approach enables household robots to recognize objects they have never seen during training and even use items as tools when appropriate, leveraging broad visual and semantic knowledge acquired from web-scale datasets.

5.2 Personal Transportation and Smart Devices

Personal transportation and smart home devices represent another frontier for VLM-enhanced robotics. Consumer drones, robotic vacuum cleaners, and lawn mowers are incorporating vision-language capabilities that enable more sophisticated autonomous operation and natural interaction with users.

Advanced applications include:

- **Intelligent camera drones**: Systems capable of interpreting high-level photographic goals such as "circle around me and capture the sunset in the background"
- **Adaptive cleaning robots**: Vacuum and lawn care systems that can identify and respond to specific obstacles or areas based on visual understanding
- **Social companion robots**: Assistive systems for elderly care that combine conversation capabilities with physical assistance tasks
- **Smart home integration**: Robotic systems that can understand and execute complex household management tasks through natural language interfaces

These developments represent constrained implementations of broader VLM capabilities, demonstrating how advanced AI technologies can be adapted for specific consumer applications while maintaining safety and usability standards.

6 Cross-Cutting Trends and Future Directions

6.1 Open-Source Robotics and Model Sharing

The democratization of advanced robotics capabilities through open-source platforms represents one of the most significant trends in the field. Hugging Face's LeRobot library exemplifies this movement by providing state-of-the-art models, demonstration datasets, and simulation tools to the broader research and development community [1].

This open-source approach creates several advantages:

- **Accelerated innovation**: Reduced barriers to entry enable more researchers and organizations to contribute to robotic capabilities
- **Data flywheel effects**: Successful policies and datasets are shared, enabling collaborative improvement of robot learning systems
- **Skill repository development**: Emergence of robot skill libraries similar to software application stores, covering diverse tasks from cooking to assembly
- **Standardization**: Common platforms and interfaces facilitate interoperability between different robotic systems and capabilities

The collaboration between Hugging Face and NVIDIA further accelerates this trend by combining open-source model development with advanced simulation and training infrastructure [26]. This partnership demonstrates how industry leaders can support community-driven development while advancing their own technological capabilities.

6.2 Simulation and Reinforcement Learning Integration

The integration of high-fidelity simulation with reinforcement learning represents a critical technological trend enabling rapid development and deployment of robot capabilities. Advanced physics simulators such as NVIDIA Isaac Sim, combined with generative AI systems, can create diverse training scenarios automatically, allowing robots to accumulate extensive experience before physical deployment.

Key technological components include:

- **Procedural scenario generation**: AI systems that create diverse training environments and edge cases automatically
- **Sim-to-real transfer**: Techniques for ensuring that behaviors learned in simulation transfer effectively to physical robots
- **Hybrid learning approaches**: Combining human demonstrations with simulation-based exploration to achieve robust policies
- **Continuous learning**: Systems that continue to improve through ongoing interaction between simulation and real-world experience

This simulation-driven approach significantly reduces development time for new robotic capabilities. Tasks that previously required months of manual programming can now be addressed through imitation learning and simulation-based training in a matter of days or weeks.

6.3 Large Language Models as Robotic Planners

Large language models are increasingly being integrated into robotic systems as high-level planners and reasoning engines, extending their capabilities beyond vision and language understanding to strategic thinking and problem-solving [32]. These systems can decompose complex goals into executable sub-tasks and provide strategic guidance when robots encounter unexpected situations.

Applications of LLMs in robotics include:

- **Task decomposition**: Breaking down complex objectives into manageable steps that robots can execute. GPT-4's code generation capabilities enable automatic translation of high-level goals into executable robot action sequences [33]
- **Strategy generation**: Providing alternative approaches when initial plans encounter obstacles or failures. Google's SayCan project demonstrates how language models can generate multiple task strategies and evaluate their feasibility in robotic contexts [16]
- **Knowledge integration**: Combining world knowledge from training data with specific task requirements and constraints. Large language models bring vast commonsense knowledge about object properties, spatial relationships, and task dependencies to robotic planning [11]

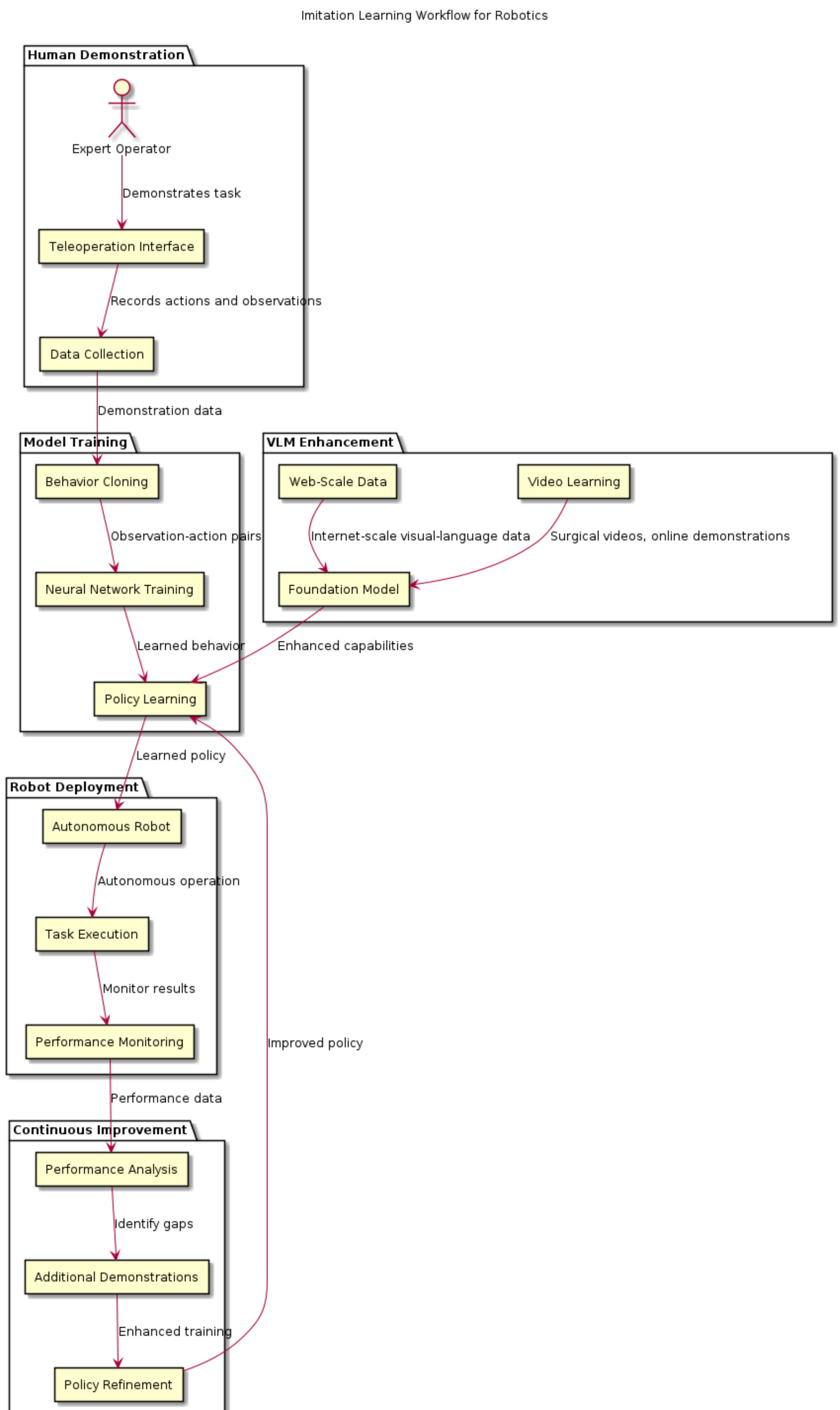

Figure 2: Comprehensive workflow for imitation learning in robotics, enhanced by VLM capabilities, incorporating best practices from Hussein et al. [14] and recent surgical robotics applications [5]

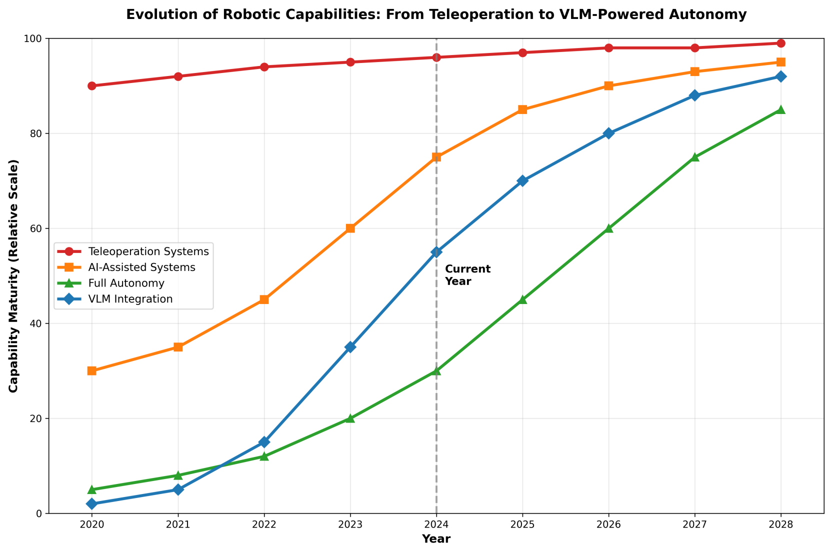

Figure 3: Timeline showing the evolution of robotic capabilities from teleoperation to VLM-powered autonomy, tracing developments from traditional systems to modern foundation models [18]

- **Design optimization**: Assisting in the development of robot morphologies and tool designs for specific applications. Recent research in LLM-guided robot design shows how language models can suggest optimal configurations for specific tasks [32]

Research has demonstrated LLM-based systems that can generate detailed plans for complex tasks, such as retrieving tools from locked storage by determining the sequence of actions required: locate storage, navigate to location, unlock container, and retrieve item. These capabilities suggest a future where robots can engage in more sophisticated reasoning and planning processes.

6.4 Challenges and Limitations

Despite significant progress, several challenges remain in the development of VLM-powered autonomous robotics:

- **Safety and reliability**: Ensuring predictable and safe behavior in unpredictable real-world environments

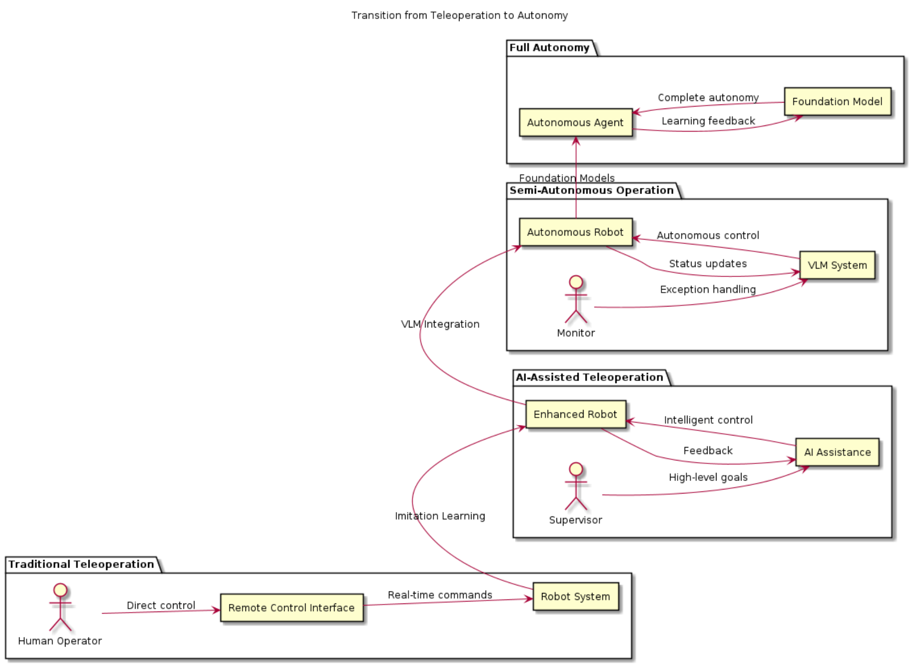

Figure 4: Evolution from traditional teleoperation to full autonomy through VLM integration, showing progression from systems like da Vinci [15] to modern AI-powered autonomous platforms

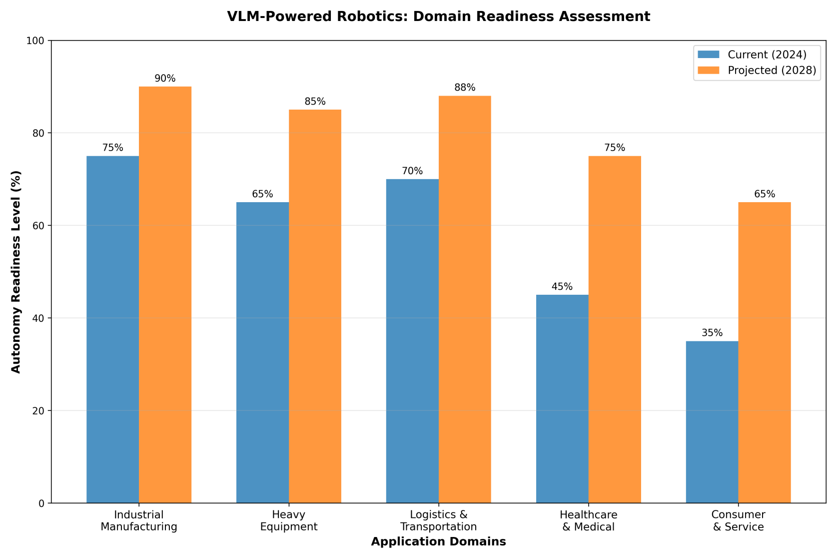

Figure 5: Current and projected readiness levels for VLM-powered robotics across different application domains, based on industry analysis and deployment patterns observed in consumer [30], industrial [19], and healthcare robotics [27]

- **Computational requirements**: Managing the significant processing power needed for real-time VLM inference on robotic platforms
- **Domain adaptation**: Transferring capabilities learned in one environment or application to different contexts
- **Human-robot interaction**: Developing intuitive and trustworthy interfaces for collaboration between humans and autonomous systems

7 Future Outlook and Research Directions

7.1 Next Four Years: Projected Developments

The next four years promise significant advances in VLM-powered robotics across multiple dimensions. Industry analysts and researchers anticipate several key developments that will shape the field:

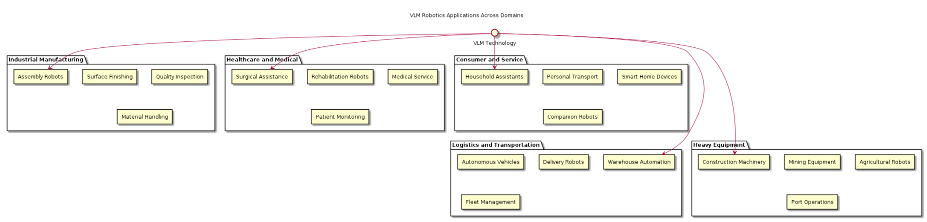

Figure 6: Overview of VLM technology applications across major robotics domains, illustrating real-world deployments from warehouse automation [24] to autonomous vehicles [6]

In industrial applications, we expect to see widespread deployment of AI-powered robotic cells that can adapt to new tasks through natural language instruction and demonstration learning. Manufacturing facilities will increasingly employ robots capable of handling high-mix production without extensive reprogramming, addressing labor shortages and increasing operational flexibility.

Healthcare robotics will likely see the introduction of semi-autonomous surgical assistants capable of performing routine procedures under human supervision. Rehabilitation and assistive robotics will benefit from improved understanding of human motion and intent, enabling more natural and effective therapeutic interventions.

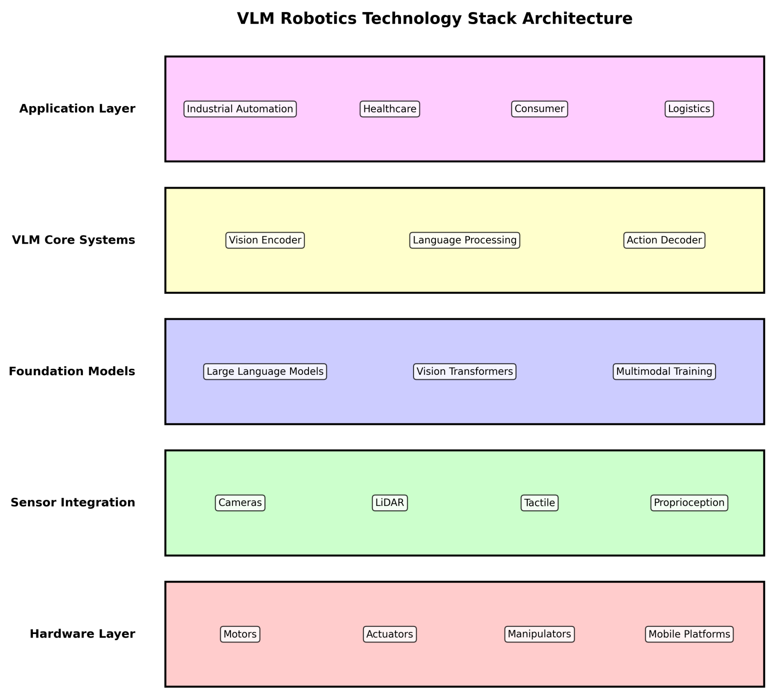

Figure 7: Technology stack architecture for VLM-powered robotic systems, incorporating components from modern multimodal foundation models [29] and simulation platforms [17]

Consumer robotics will evolve toward more capable household assistants that can understand complex requests and adapt to family routines and preferences. Personal transportation devices and smart home systems will incorporate increasingly sophisticated autonomous capabilities while maintaining safety and user control.

7.2 Long-term Vision: Toward General-Purpose Autonomy

The ultimate goal of VLM-powered robotics extends beyond domain-specific applications toward general-purpose autonomous systems capable of operating effectively across diverse environments and tasks. This vision requires continued advances in several key areas:

- **Multimodal foundation models**: Development of more powerful and efficient models that can process visual, auditory, tactile, and linguistic information simultaneously. Research directions include unified architectures that can handle multiple sensor modalities and output types, building upon successes in models like GPT-4V and Flamingo [11], [12]
- **Continual learning**: Systems that can acquire new skills and knowledge throughout their operational lifetime without catastrophic forgetting. This capability is essential for robots that must adapt to changing environments and learn from ongoing interactions with humans and objects [18]
- **Robust reasoning**: AI architectures that can handle uncertainty, ambiguity, and unexpected situations with appropriate caution and adaptability. Future systems must combine the pattern recognition capabilities of neural networks with symbolic reasoning for reliable decision-making in novel scenarios [29]
- **Ethical and social integration**: Frameworks for ensuring that autonomous robots operate in ways that align with human values and social norms. This includes developing transparent decision-making processes, ensuring equitable access to robotic assistance, and maintaining human agency in human-robot collaborations [9]

8 Conclusion

Visual language models are fundamentally transforming robotics by enabling the transition from teleoperated systems to autonomous agents capable of understanding, learning, and adapting to complex real-world environments. This survey has examined the current state of this transformation across industrial, healthcare, consumer, and logistics applications, revealing both significant progress and ongoing challenges.

The convergence of imitation learning, large language models, and multimodal AI is creating unprecedented opportunities for robotic systems to acquire complex skills through demonstration and natural language interaction. Open-source platforms and collaborative development approaches are accelerating innovation while making advanced capabilities accessible to a broader community of researchers and developers.

However, the path to general-purpose autonomous robotics requires continued advances in safety, reliability, and human-robot interaction. Success will depend on addressing fundamental challenges in reasoning under uncertainty, ensuring predictable behavior in dynamic environments, and developing trustworthy systems that augment rather than replace human capabilities.

As we look toward the future, the integration of visual language models with robotic systems promises to create a new generation of intelligent machines that can work alongside humans as capable and intuitive partners. The realization of this vision will require sustained research and development across multiple disciplines, but the potential benefits—safer workplaces, more effective healthcare, and enhanced quality of life—make this one of the most compelling frontiers in modern technology.

Acknowledgments

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